

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 14A

STRONG & WEAK ACIDS, PH SCALE, KW & KA CALCULATIONS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 14: ACID-BASE EQUILIBRIA
SUB-TOPIC:	14A: Strong & Weak Acids, pH Scale, Kw & Ka Calculations
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	226 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Brønsted-Lowry Theory & Conjugate Pairs****[3 Marks]****WCH14/01/Jan23/Q11**

According to Brønsted-Lowry theory, acid-base reactions involve proton transfer.

(a) Define a Brønsted-Lowry acid and base. [1]

(b) Identify conjugate acid-base pairs in: $\text{HNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NO}_3^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$. [2]

Q2. 2. pH Definition & Strong Acid Calculations**[4 Marks]****WCH14/01/Oct22/Q12**

Calculate the pH of the following strong acid solutions at 298 K:

(a) 0.0500 mol dm⁻³ HCl(aq). [2]

(b) 0.0250 mol dm⁻³ H₂SO₄(aq) (assume complete dissociation of both protons). [2]

Q3. 3. Ionic Product of Water Kw & Strong Base pH**[4 Marks]****WCH14/01/Jun22/Q13**

At 298 K, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

(a) Write the K_w expression and state the pH of pure water at 298 K. [2]

(b) Calculate the pH of 0.0400 mol dm⁻³ NaOH(aq) at 298 K. [2]

[3 Marks]

WCH14/01/Jan22/Q14

Q4. 4. Weak Acid Ka & pKa Definitions

Methanoic acid, HCOOH , is a weak acid with $K_a = 1.78 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K.

(a) Write the dissociation equation and K_a expression for HCOOH . [2]

(b) Calculate $\text{p}K_a = -\log_{10} K_a$ for HCOOH . [1]

[4 Marks]

WCH14/01/Oct21/Q13

Q5. 5. Weak Acid pH Calculation: Ethanoic Acid

Calculate the pH of a $0.150 \text{ mol dm}^{-3}$ solution of ethanoic acid, CH_3COOH ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K).

(a) State two assumptions made in weak acid pH calculations. [2]

(b) Calculate $[\text{H}^+]$ and the pH of the solution. [2]

[4 Marks]

WCH14/01/Jun21/Q14

Q6. 6. Temperature Effect on K_w & Pure Water pH

At 37°C (310 K), $K_w = 2.40 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$.

(a) Calculate the pH of pure water at 37°C . [2]

(b) Explain why pure water remains neutral at 37 °C despite having $\text{pH} < 7.00$. [2]

Q7. 7. Degree of Dissociation α of Weak Acids

[4 Marks]

WCH14/01/Jan21/Q15

A 0.100 mol dm^{-3} solution of a weak acid HA has $\text{pH} = 3.00$.

(a) Calculate $[\text{H}^+]$ and the degree of dissociation $\alpha = [\text{H}^+] / [\text{HA}]_0$. [2]

(b) Calculate K_a for HA. [2]

Q8. 8. Dilution Effect on Strong vs Weak Acid pH

[4 Marks]

WCH14/01/Oct20/Q13

10.0 cm^3 of 0.100 M HCl and 10.0 cm^3 of 0.100 M CH_3COOH are each diluted to 100.0 cm^3 with distilled water.

(a) Calculate the new pH of the diluted HCl solution. [2]

(b) Explain why diluting a weak acid by a factor of 10 increases its pH by less than 1.00 unit. [2]

Q9. 9. pH of Mixed Strong Acid Solutions**[4 Marks]**
WCH14/01/Jun20/Q14

50.0 cm³ of 0.200 M HCl is mixed with 50.0 cm³ of 0.100 M HNO₃.

(a) Calculate total moles of H⁺ and total volume of solution. [2]

(b) Calculate the final pH of the mixture. [2]

Q10. 10. pH of Mixed Strong Base Solutions**[4 Marks]**
WCH14/01/Jan20/Q15

25.0 cm³ of 0.100 M NaOH is mixed with 75.0 cm³ of 0.050 M KOH at 298 K ($K_w = 1.00 \times 10^{-14}$).

(a) Calculate total moles of OH⁻ and [OH⁻] in the mixture. [2]

(b) Calculate [H⁺] and final pH. [2]

Q11. 11. Calculation of K_a from Measured pH**[4 Marks]**
WCH14/01/Oct19/Q15

A 0.200 mol dm⁻³ solution of propanoic acid, C₂H₅COOH, has a measured pH of 2.89 at 298 K.

(a) Calculate [H⁺]. [1]

(b) Calculate K_a and pK_a for propanoic acid. [3]

[4 Marks]

Q12. 12. Amphoteric Behavior of Water and Amino Acids

WCH14/01/Jun19/Q16

Amphoteric substances can act as both Brønsted–Lowry acids and bases.

(a) Write two equations showing H_2O acting as a base and as an acid. [2]

(b) Show how the zwitterion of glycine, $\text{H}_3\text{N}^+\text{--CH}_2\text{--COO}^-$, acts as a buffer by accepting and donating protons. [2]

Q13. 13. pH Calculation of Strong Acid–Base Partial Neutralisation

[5 Marks]

WCH14/01/Jan19/Q17

50.0 cm³ of 0.200 M HCl is added to 25.0 cm³ of 0.300 M NaOH at 298 K.

(a) Calculate initial moles of H^+ and OH^- . [2]

(b) Determine excess H^+ or OH^- and calculate the final pH. [3]

Q14. 14. pH Calculation of Base Excess Neutralisation**[5 Marks]****WCH14/01/sample/Q14**

25.0 cm³ of 0.200 M HCl is added to 50.0 cm³ of 0.200 M NaOH at 298 K.

(a) Determine excess OH⁻ moles and [OH⁻] in final volume. [3]

(b) Calculate final pH using $K_w = 1.00 \times 10^{-14}$. [2]

Q15. 15. Comparison of Weak Acid Strengths via K_a and pK_a **[3 Marks]****WCH14/01/sample/Q15**

Acid A: $pK_a = 3.75$; Acid B: $pK_a = 4.76$; Acid C: $pK_a = 9.25$.

(a) Rank acids A, B, C in order of increasing acid strength and explain. [3]

Q16. 16. Relative Acidity of Chloro-substituted Ethanoic Acids**[4 Marks]****WCH14/01/sample/Q16**

pK_a values: CH₃COOH (4.76), CH₂ClCOOH (2.86), CHCl₂COOH (1.29), CCl₃COOH (0.65).

(a) Explain the effect of chlorine substitution on weak acid strength in terms of inductive effects. [4]

Q17. 17. pH of Monobasic vs Dibasic Weak Acids**[4 Marks]****WCH14/01/Sample/Q17**

Ethanedioic acid, $(\text{COOH})_2$, has $K_{a1} = 5.6 \times 10^{-2} \text{ mol dm}^{-3}$ and $K_{a2} = 5.4 \times 10^{-5} \text{ mol dm}^{-3}$.

(a) Explain why K_{a1} is vastly larger than K_{a2} . [2]

(b) Calculate pH of 0.100 M $(\text{COOH})_2$ considering only K_{a1} . [2]

Q18. 18. pH of Buffer-like Salt Solutions of Weak Acids**[4 Marks]****WCH14/01/Sample/Q18**

Sodium ethanoate CH_3COONa dissolves in water to give a weakly alkaline solution.

(a) Write an ionic equation for the hydrolysis of ethanoate ion in water. [2]

(b) Calculate $K_h = K_w / K_a$ and the pH of 0.100 M CH_3COONa . [2]

Q19. 19. Calculations Involving pK_w **[3 Marks]****WCH14/01/Sample/Q19**

pK_w is defined as $-\log_{10} K_w$.

(a) Show that $\text{pH} + \text{pOH} = pK_w$ at any temperature. [2]

(b) Calculate pK_w at 298 K ($K_w = 1.00 \times 10^{-14}$). [1]

Q20. 20. Exact Weak Acid Formula without Approximations

[5 Marks]

WCH14/01/Sample/Q20

When K_a is large (e.g. $K_a = 1.0 \times 10^{-2} \text{ mol dm}^{-3}$), $[HA]_{eq} = [HA]_0 - [H^+]$ cannot be approximated to $[HA]_0$.

(a) Set up the quadratic equation $K_a = [H^+]^2 / ([HA]_0 - [H^+])$ for 0.050 M acid with $K_a = 0.010$ M. [2]

(b) Solve for $[H^+]$ and calculate pH. [3]

Q21. 21. Temperature Dependence of K_a of Weak Acids

[4 Marks]

WCH14/01/Sample/Q21

Dissociation of weak acid $HA \rightleftharpoons H^+ + A^-$ is endothermic ($\Delta H > 0$).

(a) Predict how increasing temperature affects K_a and pH of a weak acid solution. [4]

Q22. 22. Acidity of Hydrated Metal Cations $[Al(H_2O)_6]^{3+}$

[4 Marks]

WCH14/01/Sample/Q22

Hexaaquaaluminum(III) ions act as a weak Brønsted-Lowry acid in aqueous solution (pH ~3).

(a) Write an equation for the dissociation of $[Al(H_2O)_6]^{3+}(aq)$. [2]

(b) Explain why Al^{3+} is acidic while Na^{+} is neutral. [2]

Q23. 23. pH Calculation of Hydrochloric Acid at Ultra-Low Concentration

[4 Marks]

WCH14/01/Sample/Q23

Calculate the pH of $1.0 \times 10^{-8} \text{ mol dm}^{-3}$ HCl(aq) at 298 K.

(a) Explain why pH is NOT equal to 8.00. [2]

(b) Account for water auto-ionization $[\text{H}^{+}]_{\text{water}}$ and calculate true pH. [2]

Q24. 24. Calculations of pKa and Ka for Phenol

[4 Marks]

WCH14/01/Sample/Q24

Phenol, $\text{C}_6\text{H}_5\text{OH}$, is a very weak acid with $K_a = 1.0 \times 10^{-10} \text{ mol dm}^{-3}$ at 298 K.

(a) Calculate pH of a $0.200 \text{ mol dm}^{-3}$ phenol solution. [3]

(b) Explain why phenol is much weaker than ethanoic acid ($K_a = 1.74 \times 10^{-5}$). [1]

Q25. 25. Relative Acidity of Alcohols, Phenols, and Carboxylic Acids**[5 Marks]****WCH14/01/Sample/Q25**

Compare pK_a values: Ethanol (16), Phenol (10), Ethanoic acid (4.76).

(a) Explain the trend in acid strength across these three organic functional groups. [5]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Multi-Step Weak Acid Neutralisation
pH Profile****[5 Marks]**
WCH14/01/Hard/Q26

25.0 cm³ of 0.100 M CH₃COOH ($K_a = 1.74 \times 10^{-5}$) is titrated with 0.100 M NaOH.

Calculate pH at: (i) $V = 0$ cm³, (ii) $V = 12.5$ cm³, (iii) $V = 25.0$ cm³.

(a) Calculate initial pH at $V = 0$ cm³. [1]

(b) Calculate half-equivalence pH at $V = 12.5$ cm³. [1]

(c) Calculate equivalence point pH at $V = 25.0$ cm³. [3]

Q27. 27. A* Challenge: pH of Polyprotic Phosphoric Acid H₃PO₄**[5 Marks]**
WCH14/01/Hard/Q27

H₃PO₄ has $K_{a1} = 7.1 \times 10^{-3}$, $K_{a2} = 6.3 \times 10^{-8}$, $K_{a3} = 4.5 \times 10^{-13}$ mol dm⁻³.

(a) Explain why $K_{a1} \gg K_{a2} \gg K_{a3}$. [2]

(b) Calculate $[H^+]$ and pH of 0.100 M H₃PO₄ using quadratic formula for K_{a1} . [3]

Q28. 28. A* Challenge: Enthalpy of Neutralisation of Strong vs Weak Acids**[5 Marks]**
WCH14/01/Hard/Q28

Delta_neut H° (kJ mol⁻¹): HCl + NaOH = -57.1; HNO₃ + KOH = -57.1; CH₃COOH + NaOH = -55.2; HCN + NaOH = -11.7.

(a) Explain why strong acid-strong base neutralisations all have Delta_neut H° = -57.1 kJ mol⁻¹. [2]

(b) Explain why Delta_neut H° for weak acids is less exothermic (e.g. HCN = -11.7 kJ mol⁻¹). [3]

Q29. 29. A* Challenge: Kw Temperature Dependence & Thermodynamic Parameters**[5 Marks]**
WCH14/01/Hard/Q29

Kw data: 298 K (1.00 x 10⁻¹⁴), 323 K (5.48 x 10⁻¹⁴), 348 K (2.34 x 10⁻¹³).

(a) Plot / calculate gradient of ln Kw vs 1/T to find Delta H° for water auto-ionization. [4]

(b) State the sign of Delta H°. [1]

Q30. 30. A* Challenge: Isoelectric Point pl of Amino Acids**[5 Marks]**
WCH14/01/Hard/Q30

Alanine has pKa1 = 2.34 (-COOH) and pKa2 = 9.69 (-NH₃⁺).

(a) Define isoelectric point pl. [1]

(b) Calculate $pI = (pK_{a1} + pK_{a2}) / 2$. [2]

(c) Draw the structure of alanine at pH 1.0, pH 6.0, and pH 12.0. [2]

Q31. 31. A* Challenge: Buffer Capacity Limits and Salt Addition

[5 Marks]
WCH14/01/Hard/Q31

1.00 dm³ solution containing 0.100 M CH₃COOH and 0.100 M CH₃COONa.

(a) Calculate initial pH. [1]

(b) Calculate new pH after adding 0.020 mol of gaseous HCl. [2]

(c) Calculate new pH after adding 0.150 mol of NaOH (exceeding buffer capacity). [2]

Q32. 32. A* Challenge: pH of Solutions of Salts of Weak Acids and Weak Bases

[5 Marks]
WCH14/01/Hard/Q32

Ammonium ethanoate, CH₃COONH₄, is formed from weak acid CH₃COOH ($K_a = 1.74 \times 10^{-5}$) and weak base NH₃ ($K_b = 1.74 \times 10^{-5}$).

(a) Show that pH of CH₃COONH₄ solution is given by $pH = \frac{1}{2} (pK_w + pK_a - pK_b)$. [3]

(b) Calculate the pH of 0.100 M $\text{CH}_3\text{COONH}_4$ at 298 K. [2]

Q33. 33. A* Challenge: Relative Acidity of Substituted Benzoic Acids

[5 Marks]
WCH14/01/Hard/Q33

pKa values: Benzoic acid $\text{C}_6\text{H}_5\text{COOH}$ (4.20), 4-nitrobenzoic acid (3.44), 4-methoxybenzoic acid (4.47).

(a) Explain the effect of electron-withdrawing ($-\text{NO}_2$) vs electron-donating ($-\text{OCH}_3$) substituents on pKa. [5]

Q34. 34. A* Challenge: Calculations of Dissociation Constant from Conductance Data

[5 Marks]
WCH14/01/Hard/Q34

Molar conductivity of 0.0100 M CH_3COOH is $16.2 \text{ S cm}^2 \text{ mol}^{-1}$; limiting molar conductivity $\Lambda_{\infty} = 390.6 \text{ S cm}^2 \text{ mol}^{-1}$.

(a) Calculate degree of dissociation $\alpha = \Lambda / \Lambda_{\infty}$. [2]

(b) Calculate K_a using Ostwald's dilution law $K_a = c \alpha^2 / (1 - \alpha)$. [3]

Q35. 35. A* Challenge: pH Calculation of Carbonic Acid - Bicarbonate System**[5 Marks]**
WCH14/01/Hard/Q35

In blood plasma: $\text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ ($K_{a1} = 4.5 \times 10^{-7} \text{ mol dm}^{-3}$, $\text{p}K_{a1} = 6.35$).

(a) Calculate ratio $[\text{HCO}_3^-] / [\text{H}_2\text{CO}_3]$ required to maintain normal blood pH = 7.40. [3]

(b) Explain how hyperventilation causes respiratory alkalosis. [2]

Q36. 36. A* Challenge: Acid Base Indicator Equivalence Point Selection**[5 Marks]**
WCH14/01/Hard/Q36

An indicator HIn is a weak acid with $K_{in} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}$ ($\text{p}K_{in} = 5.00$). HIn is yellow, In^- is red.

(a) Calculate the pH range over which the indicator changes colour ($[\text{In}^-]/[\text{HIn}]$ ratio 0.1 to 10). [3]

(b) Explain why this indicator is suitable for titrating a strong acid with a weak base. [2]

Q37. 37. A* Challenge: pH of Saturated Solutions of Sparingly Soluble Bases**[5 Marks]**
WCH14/01/Hard/Q37

For $\text{Ca}(\text{OH})_2$, solubility product $K_{\text{sp}} = 5.5 \times 10^{-6} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K.

(a) Calculate molar solubility s of $\text{Ca}(\text{OH})_2$ in mol dm^{-3} . [2]

(b) Calculate $[\text{OH}^-]$ and the pH of a saturated $\text{Ca}(\text{OH})_2$ solution. [3]

Q38. 38. A* Challenge: Acidic Properties of Non-Metal Oxides**[5 Marks]**
WCH14/01/Hard/Q38

Compare hydration reactions of P_4O_{10} , SO_3 , and Cl_2O_7 with water.

(a) Write balanced equations for the reaction of each oxide with water forming oxoacids. [3]

(b) Rank H_3PO_4 , H_2SO_4 , and HClO_4 in order of increasing acid strength with justification. [2]

Q39. 39. A* Challenge: Common Ion Effect on Weak Acid Dissociation**[5 Marks]**
WCH14/01/Hard/Q39

Calculate the pH of 0.100 M CH_3COOH ($K_{\text{a}} = 1.74 \times 10^{-5}$) when 0.050 M HCl is added.

(a) Explain using Le Chatelier's principle how added H^+ affects ethanoic acid dissociation. [2]

(b) Calculate $[\text{CH}_3\text{COO}^-]$ and overall pH of the mixture. [3]

**Q40. 40. A* Challenge: Exact Calculation of Water
Dissociation Contribution**

[5 Marks]
WCH14/01/Hard/Q40

Show that for any weak acid HA with concentration c and dissociation constant K_a , $[\text{H}^+] = \sqrt{K_a c + K_w}$.

(a) Derive this relationship considering both $\text{HA} \rightarrow \text{H}^+ + \text{A}^-$ and $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$. [5]

**Q41. 41. A* Challenge: pH Calculation of Hydrofluoric Acid HF
Solution**

[5 Marks]
WCH14/01/Hard/Q41

Hydrofluoric acid HF has $K_a = 6.8 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K.

(a) Calculate pH of 0.050 M HF using the quadratic formula. [4]

(b) Explain why HF is a weak acid despite fluorine being the most electronegative element. [1]

**Q42. 42. A* Challenge: Thermodynamic Calculation of Acid
Dissociation Constants**

[5 Marks]
WCH14/01/Hard/Q42

For CH_3COOH dissociation: $\Delta H^\circ = -0.40 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}}^\circ = -92.0 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K.

(a) Calculate $\Delta G^\circ = \Delta H^\circ - T \Delta S_{\text{system}}^\circ$. [2]

(b) Calculate K_a using $\Delta G^\circ = -RT \ln K_a$. [3]

**Q43. 43. A* Challenge: Acidity of Hydrated Iron(III) vs Iron(II)
Complexes**

[5 Marks]
WCH14/01/Hard/Q43

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ has $\text{p}K_a = 2.2$, while $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ has $\text{p}K_a = 5.9$.

(a) Explain why 0.100 M FeCl_3 solution is strongly acidic ($\text{pH} \sim 2$) while 0.100 M FeCl_2 is weakly acidic ($\text{pH} \sim 5$). [5]

**Q44. 44. A* Challenge: Fractional Ionization and Dilution Law
Proof**

[5 Marks]
WCH14/01/Hard/Q44

Ostwald's dilution law states $K_a = c \alpha^2 / (1 - \alpha)$.

(a) Show that for small α , $\alpha = \sqrt{Ka / c}$. [2]

(b) Calculate α for 1.00 M, 0.10 M, and 0.010 M CH_3COOH ($K_a = 1.74 \times 10^{-5}$). [3]

Q45. 45. A* Challenge: Dissociation of Sulfuric Acid in Concentrated Solutions

[5 Marks]

WCH14/01/Hard/Q45

H₂SO₄ undergoes two dissociation steps: H₂SO₄ → H⁺ + HSO₄⁻ (complete); HSO₄⁻ ⇌ H⁺ + SO₄²⁻ (K_{a2} = 1.2 × 10⁻²).

(a) Calculate $[H^+]$ and pH of 0.050 M H_2SO_4 considering K_{a2} via quadratic formula. [5]

Q46. 46. A* Challenge: Acidity of Hydrated Chromium(III) Complex Ions

[5 Marks]

WCH14/01/Hard/Q46

$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ has $K_a = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$ ($\text{p}K_a = 4.00$).

(a) Calculate pH of 0.050 M $\text{Cr}(\text{NO}_3)_3$ solution. [3]

(b) Write balanced equation for reaction with Na_2CO_3 producing CO_2 gas bubbles. [2]

Q47. 47. A* Challenge: Neutralisation Enthalpy of Diprotic Acids

[5 Marks]
WCH14/01/Hard/Q47

Neutralisation of 1 mole of H_2SO_4 with 2 moles of NaOH releases 114.2 kJ.

(a) Calculate the standard enthalpy of neutralisation per mole of water formed. [2]

(b) Neutralisation of 1 mole of ethanedioic acid $(\text{COOH})_2$ with 2 moles of NaOH releases 106.4 kJ. Calculate heat absorbed per mole for weak acid dissociation. [3]

Q48. 48. A* Challenge: pH Calculation of Ammonium Chloride Salt Solution

[5 Marks]
WCH14/01/Hard/Q48

Calculate the pH of 0.200 mol dm^{-3} NH_4Cl solution at 298 K (K_b for $\text{NH}_3 = 1.74 \times 10^{-5}$, $K_w = 1.00 \times 10^{-14}$).

(a) Calculate K_a for conjugate acid $\text{NH}_4^+ = K_w / K_b$. [2]

(b) Calculate $[\text{H}^+]$ and pH of 0.200 M NH_4Cl . [3]

Q49. 49. A* Challenge: Determination of Weak Acid K_a from Titration Half-Way Point**[5 Marks]**
WCH14/01/Hard/Q49

A 25.0 cm³ sample of an unknown monoprotic weak acid HA is titrated with 0.100 M NaOH. Equivalence point reached after adding 18.40 cm³ of NaOH. At $V = 9.20$ cm³, pH = 3.85.

(a) Deduce p K_a and K_a of the weak acid. [2]

(b) Calculate the concentration of original weak acid solution. [3]

Q50. 50. A* Challenge: Complete Acid-Base Master Synthesis**[6 Marks]**
WCH14/01/Hard/Q50

A solution is prepared by dissolving 0.050 mol of weak acid HA (p K_a = 4.20) in 250 cm³ of water. (K_w = 1.00×10^{-14} at 298 K)

(a) Calculate concentration [HA]₀ and K_a . [2]

(b) Calculate pH of the initial solution. [2]

(c) Calculate pH after adding 0.010 mol of solid NaOH. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 14A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Brønsted–Lowry Theory & Conjugate Pairs 1. (a) Brønsted–Lowry acid is a proton donor; base is a proton acceptor (1). 1. (b) Pair 1: Acid HNO_3 / Base NO_3^- (1); Pair 2: Base H_2O / Acid H_3O^+ (1).	[3]
Q2	Q2: 2. pH Definition & Strong Acid Calculations 2. (a) $[\text{H}^+] = 0.0500 \text{ M} \Rightarrow \text{pH} = -\log_{10}(0.0500) = 1.30$ (2). 2. (b) $[\text{H}^+] = 2 \times 0.0250 = 0.0500 \text{ M} \Rightarrow \text{pH} = -\log_{10}(0.0500) = 1.30$ (2).	[4]
Q3	Q3: 3. Ionic Product of Water K_w & Strong Base pH 3. (a) $K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$ (1). Pure water $[\text{H}^+] = 1.00 \times 10^{-7} \text{ M} \Rightarrow \text{pH} = 7.00$ (1). 3. (b) $[\text{OH}^-] = 0.0400 \text{ M} \Rightarrow [\text{H}^+] = 1.00 \times 10^{-14} / 0.0400 = 2.50 \times 10^{-13} \text{ M} \Rightarrow \text{pH} = -\log_{10}(2.50 \times 10^{-13}) = 12.60$ (2).	[4]
Q4	Q4: 4. Weak Acid K_a & $\text{p}K_a$ Definitions 4. (a) $\text{HCOOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCOO}^-(\text{aq})$; $K_a = [\text{H}^+][\text{HCOO}^-] / [\text{HCOOH}]$ (2). 4. (b) $\text{p}K_a = -\log_{10}(1.78 \times 10^{-4}) = 3.75$ (1).	[3]
Q5	Q5: 5. Weak Acid pH Calculation: Ethanoic Acid 5. (a) 1. $[\text{H}^+] = [\text{CH}_3\text{COO}^-]$ (dissociation of H_2O is negligible) (1). 2. $[\text{CH}_3\text{COOH}]_{\text{eq}} \sim [\text{CH}_3\text{COOH}]_{\text{initial}}$ (ionization is negligible) (1). 5. (b) $[\text{H}^+] = \sqrt{K_a \times [\text{HA}]} = \sqrt{1.74 \times 10^{-5} \times 0.150} = \sqrt{2.61 \times 10^{-6}} = 1.616 \times 10^{-3} \text{ M} \Rightarrow \text{pH} = 2.79$ (2).	[4]
Q6	Q6: 6. Temperature Effect on K_w & Pure Water pH 6. (a) $[\text{H}^+] = \sqrt{K_w} = \sqrt{2.40 \times 10^{-14}} = 1.549 \times 10^{-7} \text{ M} \Rightarrow \text{pH} = 6.81$ (2). 6. (b) Water is neutral because $[\text{H}^+] = [\text{OH}^-]$ (1). Dissociation $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ is endothermic, so K_w increases with temperature (1).	[4]
Q7	Q7: 7. Degree of Dissociation α of Weak Acids 7. (a) $[\text{H}^+] = 10^{-3.00} = 1.00 \times 10^{-3} \text{ M}$. $\alpha = 1.00 \times 10^{-3} / 0.100 = 0.010$ (1.0% ionized) (2). 7. (b) $K_a = [\text{H}^+]^2 / [\text{HA}]_0 = (1.00 \times 10^{-3})^2 / 0.100 = 1.00 \times 10^{-5} \text{ mol dm}^{-3}$ (2).	[4]
Q8	Q8: 8. Dilution Effect on Strong vs Weak Acid pH 8. (a) Dilution factor 10x $\Rightarrow [\text{HCl}] = 0.0100 \text{ M} \Rightarrow \text{pH} = 2.00$ (increases by exactly 1.00 unit) (2). 8. (b) Dilution shifts weak acid equilibrium $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ to the right (Le Chatelier), producing extra H^+ ions so $[\text{H}^+]$ decreases by less than 10x (2).	[4]
Q9	Q9: 9. pH of Mixed Strong Acid Solutions 9. (a) Moles H^+ from $\text{HCl} = 0.050 \times 0.200 = 0.010 \text{ mol}$; Moles H^+ from $\text{HNO}_3 = 0.050 \times 0.100 = 0.005 \text{ mol}$. Total $\text{H}^+ = 0.015 \text{ mol}$; Total $V = 0.100 \text{ dm}^3$ (2). 9. (b) $[\text{H}^+] = 0.015 / 0.100 = 0.150 \text{ M} \Rightarrow \text{pH} = -\log_{10}(0.150) = 0.82$ (2).	[4]
Q10	Q10: 10. pH of Mixed Strong Base Solutions 10. (a) Moles $\text{OH}^- = (0.025 \times 0.100) + (0.075 \times 0.050) = 0.0025 + 0.00375 = 0.00625 \text{ mol}$. $[\text{OH}^-] = 0.00625 / 0.100 = 0.0625 \text{ M}$ (2). 10. (b) $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0625 = 1.60 \times 10^{-13} \text{ M} \Rightarrow \text{pH} = 12.80$ (2).	[4]
Q11	Q11: 11. Calculation of K_a from Measured pH 11. (a) $[\text{H}^+] = 10^{-2.89} = 1.288 \times 10^{-3} \text{ M}$ (1). 11. (b) $K_a = (1.288 \times 10^{-3})^2 / 0.200 = 1.66 \times 10^{-5} \text{ mol dm}^{-3}$ (2). $\text{p}K_a = -\log_{10}(1.66 \times 10^{-5}) = 4.78$ (1).	[4]
Q12	Q12: 12. Amphoteric Behavior of Water and Amino Acids 12. (a) As base: $\text{H}_2\text{O} + \text{HCl} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$ (1). As acid: $\text{H}_2\text{O} + \text{NH}_3 \rightarrow \text{OH}^- + \text{NH}_4^+$ (1). 12. (b) In acid: $-\text{COO}^- + \text{H}^+ \rightarrow -\text{COOH}$ (1). In base: $-\text{NH}_3^+ + \text{OH}^- \rightarrow -\text{NH}_2 + \text{H}_2\text{O}$ (1).	[4]
Q13	Q13: 13. pH Calculation of Strong Acid–Base Partial Neutralisation 13. (a) Moles $\text{H}^+ = 0.050 \times 0.200 = 0.010 \text{ mol}$; Moles $\text{OH}^- = 0.025 \times 0.300 = 0.0075 \text{ mol}$ (2). 13. (b) Excess $\text{H}^+ = 0.010 - 0.0075 = 0.0025 \text{ mol}$ (1). Total $V = 0.075 \text{ dm}^3 \Rightarrow [\text{H}^+] = 0.0025 / 0.075 = 0.0333 \text{ M}$ (1). $\text{pH} = -\log_{10}(0.0333) = 1.48$ (1).	[5]

Q14	Q14: 14. pH Calculation of Base Excess Neutralisation 14. (a) Moles H^+ = 0.0050 mol; Moles OH^- = 0.0100 mol. Excess OH^- = 0.0050 mol in 0.075 dm ³ $\Rightarrow [OH^-]$ = 0.0667 M (3). 14. (b) $[H^+] = 1.00 \times 10^{-14} / 0.0667 = 1.50 \times 10^{-13}$ M $\Rightarrow$ pH = 12.82 (2).	[5]
Q15	Q15: 15. Comparison of Weak Acid Strengths via K_a and pK_a 15. (a) Acid C < Acid B < Acid A (1). Lower pK_a value corresponds to larger K_a value (1), indicating greater extent of dissociation and stronger acid (1).	[3]
Q16	Q16: 16. Relative Acidity of Chloro-substituted Ethanoic Acids 16. (a) Chlorine is highly electronegative and exerts an electron-withdrawing inductive effect (-I effect) (1). Withdraws electron density along C-C and C-O bonds towards chlorine atoms (1). Stabilizes the carboxylate anion ($-COO^-$) by delocalising negative charge (1). Position of equilibrium shifts right, increasing K_a and acid strength (1).	[4]
Q17	Q17: 17. pH of Monobasic vs Dibasic Weak Acids 17. (a) Removing a positive proton H^+ from a neutral molecule $(COOH)_2$ is much easier than removing H^+ from a negatively charged anion $HOOC-COO^-$ (2). 17. (b) $[H^+] = \sqrt{5.6 \times 10^{-2} \times 0.100} = \sqrt{0.0056} = 0.0748$ M $\Rightarrow$ pH = 1.13 (2).	[4]
Q18	Q18: 18. pH of Buffer-like Salt Solutions of Weak Acids 18. (a) $CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$ (2). 18. (b) $K_h = 1.00 \times 10^{-14} / 1.74 \times 10^{-5} = 5.75 \times 10^{-10}$. $[OH^-] = \sqrt{5.75 \times 10^{-10} \times 0.100} = 7.58 \times 10^{-6}$ M $\Rightarrow$ pOH = 5.12 $\Rightarrow$ pH = 8.88 (2).	[4]
Q19	Q19: 19. Calculations Involving pK_w 19. (a) $K_w = [H^+][OH^-]$. Take $-\log_{10}$ of both sides: $-\log_{10} K_w = -\log_{10}([H^+][OH^-]) = -\log_{10}[H^+] + (-\log_{10}[OH^-]) \Rightarrow pK_w = pH + pOH$ (2). 19. (b) $pK_w = 14.00$ (1).	[3]
Q20	Q20: 20. Exact Weak Acid Formula without Approximations 20. (a) $0.010 = [H^+]^2 / (0.050 - [H^+]) \Rightarrow [H^+]^2 + 0.010[H^+] - 0.00050 = 0$ (2). 20. (b) $[H^+] = (-0.010 + \sqrt{0.00010 + 0.0020}) / 2 = (-0.010 + 0.0458) / 2 = 0.0179$ M $\Rightarrow$ pH = 1.75 (3).	[5]
Q21	Q21: 21. Temperature Dependence of K_a of Weak Acids 21. (a) Endothermic dissociation ($\Delta H > 0$) means increasing temperature shifts equilibrium right to absorb heat (1). Concentration of H^+ increases, so K_a increases (1) and pK_a decreases (1). Higher $[H^+]$ means pH decreases (acid becomes more ionized at higher T) (1).	[4]
Q22	Q22: 22. Acidity of Hydrated Metal Cations $[Al(H_2O)_6]^{3+}$ 22. (a) $[Al(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$ (2). 22. (b) Al^{3+} has high charge (+3) and small ionic radius (high charge density), strongly polarizing O-H bonds of water ligands (1). Na^+ (+1, larger radius) has low charge density and cannot polarize O-H bonds (1).	[4]
Q23	Q23: 23. pH Calculation of Hydrochloric Acid at Ultra-Low Concentration 23. (a) HCl is an acid; adding acid to water cannot make the solution alkaline (pH > 7) (2). 23. (b) Total $[H^+] = [H^+]_{HCl} + [H^+]_{water} = 1.0 \times 10^{-8} + x$ where $(1.0 \times 10^{-8} + x)x = 1.0 \times 10^{-14} \Rightarrow x = 9.5 \times 10^{-8} \Rightarrow$ Total $[H^+] = 1.05 \times 10^{-7}$ M $\Rightarrow$ pH = 6.98 (2).	[4]
Q24	Q24: 24. Calculations of pK_a and K_a for Phenol 24. (a) $[H^+] = \sqrt{1.0 \times 10^{-10} \times 0.200} = \sqrt{2.0 \times 10^{-11}} = 4.47 \times 10^{-6}$ M $\Rightarrow$ pH = 5.35 (3). 24. (b) Phenoxide anion negative charge is delocalised into aromatic ring, but O-H bond in phenol is less polar than in carboxyl group (1).	[4]
Q25	Q25: 25. Relative Acidity of Alcohols, Phenols, and Carboxylic Acids 25. (a) Ethanol: Alkyl group is electron-donating (+I effect), destabilising ethoxide ion $C_2H_5O^-$ (weakest acid) (1.5). Phenol: Phenoxide oxygen lone pair delocalises into pi system of benzene ring, stabilizing anion (intermediate acid) (1.5). Ethanoic acid: Negative charge on carboxylate anion CH_3COO^- is delocalized equally over TWO highly electronegative oxygen atoms via resonance, providing maximum anion stability (strongest acid) (2).	[5]

Q26	Q26: 26. A* Challenge: Multi-Step Weak Acid Neutralisation pH Profile 26. (a) $[H^+] = \sqrt{1.74 \times 10^{-5} \times 0.100} = 1.319 \times 10^{-3} \Rightarrow pH = 2.88$ (1). 26. (b) Half-equivalence: $pH = pK_a = -\log_{10}(1.74 \times 10^{-5}) = 4.76$ (1). 26. (c) Equivalence: $[CH_3COO^-] = 0.0025 / 0.050 = 0.050$ M. $K_h = 1.00 \times 10^{-14} / 1.74 \times 10^{-5} = 5.75 \times 10^{-10}$. $[OH^-] = \sqrt{5.75 \times 10^{-10} \times 0.050} = 5.36 \times 10^{-6}$ M $\Rightarrow pOH = 5.27 \Rightarrow pH = 8.73$ (3).	[5]
Q27	Q27: 27. A* Challenge: pH of Polyprotic Phosphoric Acid H_3PO_4 27. (a) Removing positive H^+ from neutral H_3PO_4 is much easier than from 1^- ion ($H_2PO_4^-$) or 2^- ion (HPO_4^{2-}) due to increasing electrostatic attraction (2). 27. (b) $7.1 \times 10^{-3} = [H^+]^2 / (0.100 - [H^+]) \Rightarrow [H^+]^2 + 0.0071[H^+] - 0.00071 = 0 \Rightarrow [H^+] = 0.0234$ M $\Rightarrow pH = 1.63$ (3).	[5]
Q28	Q28: 28. A* Challenge: Enthalpy of Neutralisation of Strong vs Weak Acids 28. (a) All strong acids/bases are 100% ionized; essential net reaction is always $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ (2). 28. (b) Weak acids are only partially ionized. Part of the heat released during neutralisation is absorbed (endothermic) to dissociate the remaining un-ionized weak acid molecules (3).	[5]
Q29	Q29: 29. A* Challenge: K_w Temperature Dependence & Thermodynamic Parameters 29. (a) In K_w at 298 K = -32.236; at 348 K = -29.085. $(1/348 - 1/298) = -4.823 \times 10^{-4}$ K ⁻¹ . Gradient = $(-29.085 - (-32.236)) / (-4.823 \times 10^{-4}) = -6533$ K (3). $\Delta H = 6533 \times 8.31 = +54.3$ kJ mol ⁻¹ (1). 29. (b) Endothermic ($\Delta H > 0$) (1).	[5]
Q30	Q30: 30. A* Challenge: Isoelectric Point pI of Amino Acids 30. (a) pH at which an amino acid exists predominantly as a dipolar zwitterion with zero net electrical charge (1). 30. (b) $pI = (2.34 + 9.69) / 2 = 6.02$ (2). 30. (c) At pH 1: $H_3N^+-CH(CH_3)-COOH$ (fully protonated) (0.7). At pH 6: $H_3N^+-CH(CH_3)-COO^-$ (zwitterion) (0.7). At pH 12: $H_2N-CH(CH_3)-COO^-$ (fully deprotonated) (0.6).	[5]
Q31	Q31: 31. A* Challenge: Buffer Capacity Limits and Salt Addition 31. (a) $pH = pK_a = 4.76$ (1). 31. (b) Added H^+ reacts with CH_3COO^- : $[CH_3COOH] = 0.120$ M, $[CH_3COO^-] = 0.080$ M $\Rightarrow pH = 4.76 + \log_{10}(0.080/0.120) = 4.58$ (2). 31. (c) 0.150 mol OH^- neutralises all 0.100 mol CH_3COOH with 0.050 mol excess OH^- . $[OH^-] = 0.050$ M $\Rightarrow pOH = 1.30 \Rightarrow pH = 12.70$ (buffer destroyed) (2).	[5]
Q32	Q32: 32. A* Challenge: pH of Solutions of Salts of Weak Acids and Weak Bases 32. (a) Both cation NH_4^+ and anion CH_3COO^- undergo hydrolysis (1). Since $K_a = K_b$, $[H^+]$ produced by NH_4^+ equals $[OH^-]$ produced by CH_3COO^- (2). 32. (b) $pH = 1/2 (14.00 + 4.76 - 4.76) = 7.00$ (neutral salt solution) (2).	[5]
Q33	Q33: 33. A* Challenge: Relative Acidity of Substituted Benzoic Acids 33. (a) Nitro group ($-NO_2$) is strongly electron-withdrawing ($-I$ and $-M$ resonance effects) (1). Withdraws electron density from benzene ring and carboxylate group, stabilizing $Ar-COO^-$ anion (1). Increases K_a , lowering pK_a to 3.44 (stronger acid) (1). Methoxy group ($-OCH_3$) donates oxygen lone pair into aromatic π system ($+M$ effect) (1). Increases electron density on $-COO^-$, destabilising anion and decreasing acid strength ($pK_a = 4.47$) (1).	[5]
Q34	Q34: 34. A* Challenge: Calculations of Dissociation Constant from Conductance Data 34. (a) $\alpha = 16.2 / 390.6 = 0.04147$ (4.15% ionized) (2). 34. (b) $K_a = 0.0100 \times (0.04147)^2 / (1 - 0.04147) = 1.72 \times 10^{-5} / 0.9585 = 1.79 \times 10^{-5}$ mol dm ⁻³ (3).	[5]
Q35	Q35: 35. A* Challenge: pH Calculation of Carbonic Acid - Bicarbonate System 35. (a) $pH = pK_a + \log_{10}([HCO_3^-]/[H_2CO_3]) \Rightarrow 7.40 = 6.35 + \log_{10}(\text{ratio})$ (1). $\log_{10}(\text{ratio}) = 1.05$ (1). Ratio = $10^{1.05} = 11.2$ (3). 35. (b) Rapid breathing expels CO_2 gas, shifting $CO_2(aq) + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$ to the left, decreasing $[H^+]$ and raising blood pH above 7.45 (alkalosis) (2).	[5]

Q36	Q36: 36. A* Challenge: Acid Base Indicator Equivalence Point Selection 36. (a) At ratio 0.1: $\text{pH} = 5.00 + \log(0.1) = 4.00$ (1). At ratio 10: $\text{pH} = 5.00 + \log(10) = 6.00$ (1). Colour transition range = pH 4.0 to 6.0 (1). 36. (b) Strong acid-weak base titration has an equivalence point in the acidic region (pH 4–6), coinciding perfectly with this indicator's transition range (2).	[5]
Q37	Q37: 37. A* Challenge: pH of Saturated Solutions of Sparingly Soluble Bases 37. (a) $\text{Ca(OH)}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+} + 2\text{OH}^-$. $K_{\text{sp}} = s \times (2s)^2 = 4s^3 = 5.5 \times 10^{-6} \Rightarrow s^3 = 1.375 \times 10^{-6} \Rightarrow s = 0.011 \text{ mol dm}^{-3}$ (2). 37. (b) $[\text{OH}^-] = 2s = 0.0222 \text{ M}$ (1). $[\text{H}^+] = 1.00 \times 10^{-14} / 0.0222 = 4.50 \times 10^{-13} \text{ M} \Rightarrow \text{pH} = 12.35$ (2).	[5]
Q38	Q38: 38. A* Challenge: Acidic Properties of Non-Metal Oxides 38. (a) $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ (1); $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$ (1); $\text{Cl}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{HClO}_4$ (1). 38. (b) $\text{H}_3\text{PO}_4 < \text{H}_2\text{SO}_4 < \text{HClO}_4$ (1). As oxidation state of central atom increases (+5 to +6 to +7) and electronegativity increases, central atom pulls electron density away from O–H bonds, weakening O–H bond and releasing H^+ more readily (1).	[5]
Q39	Q39: 39. A* Challenge: Common Ion Effect on Weak Acid Dissociation 39. (a) Added H^+ from strong HCl increases $[\text{H}^+]$, shifting equilibrium $\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$ to the left (common ion effect), suppressing weak acid ionization (2). 39. (b) $[\text{H}^+] \sim 0.050 \text{ M} \Rightarrow \text{pH} = -\log(0.050) = 1.30$ (1). $1.74 \times 10^{-5} = (0.050 \times [\text{CH}_3\text{COO}^-]) / 0.100 \Rightarrow [\text{CH}_3\text{COO}^-] = 3.48 \times 10^{-5} \text{ mol dm}^{-3}$ (2).	[5]
Q40	Q40: 40. A* Challenge: Exact Calculation of Water Dissociation Contribution 40. (a) Charge balance: $[\text{H}^+] = [\text{A}^-] + [\text{OH}^-]$ (1). From K_{a} : $[\text{A}^-] = K_{\text{a}} [\text{HA}] / [\text{H}^+] \sim K_{\text{a}} c / [\text{H}^+]$ (2). From K_{w} : $[\text{OH}^-] = K_{\text{w}} / [\text{H}^+]$ (1). Substitute: $[\text{H}^+] = K_{\text{a}} c / [\text{H}^+] + K_{\text{w}} / [\text{H}^+] \Rightarrow [\text{H}^+]^2 = K_{\text{a}} c + K_{\text{w}} \Rightarrow [\text{H}^+] = \sqrt{K_{\text{a}} c + K_{\text{w}}}$ (1).	[5]
Q41	Q41: 41. A* Challenge: pH Calculation of Hydrofluoric Acid HF Solution 41. (a) $6.8 \times 10^{-4} = [\text{H}^+]^2 / (0.050 - [\text{H}^+]) \Rightarrow [\text{H}^+]^2 + 0.00068[\text{H}^+] - 0.000034 = 0 \Rightarrow [\text{H}^+] = 0.00550 \text{ M} \Rightarrow \text{pH} = 2.26$ (4). 41. (b) H–F bond enthalpy is exceptionally high (567 kJ mol^{-1}), requiring huge energy to break H–F bond (1).	[5]
Q42	Q42: 42. A* Challenge: Thermodynamic Calculation of Acid Dissociation Constants 42. (a) $\Delta G^\circ = -400 - 298(-92.0) = -400 + 27416 = +27016 \text{ J mol}^{-1} = +27.02 \text{ kJ mol}^{-1}$ (2). 42. (b) $\ln K_{\text{a}} = -27016 / (8.31 \times 298) = -27016 / 2476.38 = -10.9095 \Rightarrow K_{\text{a}} = e^{-10.9095} = 1.83 \times 10^{-5} \text{ mol dm}^{-3}$ (3).	[5]
Q43	Q43: 43. A* Challenge: Acidity of Hydrated Iron(III) vs Iron(II) Complexes 43. (a) Fe^{3+} has higher charge (+3 vs +2) and smaller ionic radius than Fe^{2+} (1). Fe^{3+} has vastly higher charge density, pulling electron density strongly from oxygen atoms of coordinated H_2O ligands (2). Weakens O–H bonds of H_2O ligands significantly, releasing H^+ readily into solution: $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+} + \text{H}_3\text{O}^+$ (2).	[5]
Q44	Q44: 44. A* Challenge: Fractional Ionization and Dilution Law Proof 44. (a) If $\alpha \ll 1$, $(1 - \alpha) \sim 1 \Rightarrow K_{\text{a}} \sim c \alpha^2 \Rightarrow \alpha = \sqrt{K_{\text{a}} / c}$ (2). 44. (b) At 1.00 M: $\alpha = \sqrt{1.74 \times 10^{-5} / 1.00} = 0.00417$ (0.42%) (1). At 0.10 M: $\alpha = \sqrt{1.74 \times 10^{-5} / 0.10} = 0.0132$ (1.32%) (1). At 0.010 M: $\alpha = \sqrt{1.74 \times 10^{-5} / 0.010} = 0.0417$ (4.17%) (1).	[5]
Q45	Q45: 45. A* Challenge: Dissociation of Sulfuric Acid in Concentrated Solutions 45. (a) Step 1 gives $[\text{H}^+] = 0.050 \text{ M}$, $[\text{HSO}_4^-] = 0.050 \text{ M}$. Step 2: $\text{HSO}_4^- \rightleftharpoons \text{H}^+ + \text{SO}_4^{2-}$. $K_{\text{a}2} = (0.050 + x)x / (0.050 - x) = 0.012 \Rightarrow x^2 + 0.062x - 0.00060 = 0$ (2). $x = (-0.062 + \sqrt{0.003844 + 0.00240}) / 2 = (-0.062 + 0.0790) / 2 = 0.0085 \text{ M}$ (2). Total $[\text{H}^+] = 0.050 + 0.0085 = 0.0585 \text{ M} \Rightarrow \text{pH} = 1.23$ (1).	[5]
Q46	Q46: 46. A* Challenge: Acidity of Hydrated Chromium(III) Complex Ions 46. (a) $[\text{H}^+] = \sqrt{1.0 \times 10^{-4} \times 0.050} = \sqrt{5.0 \times 10^{-6}} = 2.236 \times 10^{-3} \text{ M} \Rightarrow \text{pH} = 2.65$ (3). 46. (b) $2[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + 3\text{CO}_3^{2-} \rightarrow 2[\text{Cr}(\text{H}_2\text{O})_3(\text{OH})_3](\text{s}) + 3\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$ (2).	[5]
Q47	Q47: 47. A* Challenge: Neutralisation Enthalpy of Diprotic Acids 47. (a) $\Delta_{\text{neut}} \text{H} = -114.2 / 2 = -57.1 \text{ kJ mol}^{-1}$ (per mole of H_2O) (2). 47. (b) Theoretical heat for 2 mol $\text{H}_2\text{O} = 2 \times (-57.1) = -114.2 \text{ kJ}$. Actual heat = -106.4 kJ . Difference = $114.2 - 106.4 = +7.8 \text{ kJ}$ for 1 mol $(\text{COOH})_2 \Rightarrow +3.9 \text{ kJ mol}^{-1}$ of H^+ dissociated (3).	[5]

Q48	Q48: 48. A* Challenge: pH Calculation of Ammonium Chloride Salt Solution 48. (a) $K_a(\text{NH}_4^+) = 1.00 \times 10^{-14} / 1.74 \times 10^{-5} = 5.75 \times 10^{-10} \text{ mol dm}^{-3}$ (2). 48. (b) $[\text{H}^+] = \sqrt{K_a \times [\text{NH}_4^+]} = \sqrt{5.75 \times 10^{-10} \times 0.200} = \sqrt{1.15 \times 10^{-10}} = 1.072 \times 10^{-5} \text{ M} \Rightarrow \text{pH} = 4.97$ (3).	[5]
Q49	Q49: 49. A* Challenge: Determination of Weak Acid K_a from Titration Half-Way Point 49. (a) $V = 9.20 \text{ cm}^3$ is exact half-equivalence point ($18.40 / 2$). At half-equivalence, $[\text{HA}] = [\text{A}^-] \Rightarrow \text{pH} = \text{p}K_a = 3.85$. $K_a = 10^{-3.85} = 1.41 \times 10^{-4} \text{ mol dm}^{-3}$ (2). 49. (b) Moles acid = Moles NaOH at eq = $0.01840 \times 0.100 = 1.84 \times 10^{-3} \text{ mol}$. $[\text{HA}]_0 = 1.84 \times 10^{-3} / 0.0250 = 0.0736 \text{ mol dm}^{-3}$ (3).	[5]
Q50	Q50: 50. A* Challenge: Complete Acid-Base Master Synthesis 50. (a) $[\text{HA}]_0 = 0.050 / 0.250 = 0.200 \text{ M}$ (1). $K_a = 10^{-4.20} = 6.31 \times 10^{-5} \text{ mol dm}^{-3}$ (1). 50. (b) $[\text{H}^+] = \sqrt{6.31 \times 10^{-5} \times 0.200} = 3.55 \times 10^{-3} \text{ M} \Rightarrow \text{pH} = 2.45$ (2). 50. (c) Added OH^- neutralises 0.010 mol HA : Moles HA remaining = $0.050 - 0.010 = 0.040 \text{ mol}$; Moles A^- formed = 0.010 mol . $\text{pH} = \text{p}K_a + \log_{10}(0.010/0.040) = 4.20 + \log_{10}(0.25) = 4.20 - 0.60 = 3.60$ (2).	[6]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 14A

FAQ #1: Definition of Conjugate Acid–Base Pairs

Category: Definition Trap

COMMON EXAMINER TRAP / MISTAKE:

Confusing conjugate acid with conjugate base in equilibria.

MODEL ANSWER / EXAMINER EXPECTATION:

A conjugate acid has ONE MORE H^+ proton than its conjugate base (e.g. H_3O^+ is conjugate acid of H_2O ; NO_3^- is conjugate base of HNO_3).

FAQ #2: Sulfuric Acid H_2SO_4 Proton Count

Category: Strong Acid Calculation

COMMON EXAMINER TRAP / MISTAKE:

Treating H_2SO_4 as monobasic ($[H^+] = [acid]$) instead of dibasic ($[H^+] = 2 \times [acid]$).

MODEL ANSWER / EXAMINER EXPECTATION:

For strong dibasic acids like H_2SO_4 , 1 mole releases 2 moles of H^+ ions upon complete dissociation ($[H^+] = 2 \times c$).

FAQ #3: K_w Value Temperature DependenceCategory: K_w Temperature Effect**COMMON EXAMINER TRAP / MISTAKE:**

Assuming K_w is always 1.00×10^{-14} at all temperatures.

MODEL ANSWER / EXAMINER EXPECTATION:

K_w increases with temperature because water auto-ionization ($H_2O \rightleftharpoons H^+ + OH^-$) is ENDOTHERMIC. At $37^\circ C$, $K_w = 2.4 \times 10^{-14}$ and pure water $pH = 6.81$ (still neutral because $[H^+] = [OH^-]$).

FAQ #4: Weak Acid Approximations

Category: K_a Calculation Trap**COMMON EXAMINER TRAP / MISTAKE:**

Forgetting the two key assumptions in weak acid pH calculations.

MODEL ANSWER / EXAMINER EXPECTATION:

Assumption 1: $[H^+] = [A^-]$ (neglect H^+ from H_2O). Assumption 2: $[HA]_{eq} \sim [HA]_{initial}$ (ionization is negligible $< 5\%$).

FAQ #5: pK_a vs Acid Strength

Category: Inverse Log Scale

COMMON EXAMINER TRAP / MISTAKE:

Thinking a higher pK_a value means a stronger acid.

MODEL ANSWER / EXAMINER EXPECTATION:

$pK_a = -\log_{10} K_a$. SMALLER or more negative pK_a corresponds to LARGER K_a and a STRONGER acid.

FAQ #6: Diluting Weak Acids pH Effect

Category: Le Chatelier Dilution

COMMON EXAMINER TRAP / MISTAKE:

Assuming 10-fold dilution of weak acid increases pH by exactly 1.00 unit.

MODEL ANSWER / EXAMINER EXPECTATION:

Diluting weak acid shifts equilibrium $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ to the right, producing extra H^+ ions. 10x dilution increases pH by LESS than 1.00 unit (typically ~0.5 units).

FAQ #7: pH of Very Dilute Strong Acids**Category: Low Concentration Trap****COMMON EXAMINER TRAP / MISTAKE:**

Calculating pH of 10^{-8} M HCl as 8.00.

MODEL ANSWER / EXAMINER EXPECTATION:

An acid solution CANNOT have $\text{pH} > 7.00$. At concentrations $< 10^{-6}$ M, water auto-ionization contribution MUST be included ($[\text{H}^+] = [\text{acid}] + [\text{H}^+]_{\text{water}} \sim 1.05 \times 10^{-7} \text{ M} \Rightarrow \text{pH} = 6.98$).

FAQ #8: Inductive Effects on Weak Acid Strength**Category: Organic Acidity****COMMON EXAMINER TRAP / MISTAKE:**

Confusing electron-withdrawing vs electron-donating substituent effects.

MODEL ANSWER / EXAMINER EXPECTATION:

Electron-withdrawing groups ($-\text{Cl}$, $-\text{NO}_2$) pull electron density away from $-\text{COO}^-$, STABILIZING carboxylate anion and INCREASING acid strength (lower pK_a).

FAQ #9: pH of Salt Solutions of Weak Acids**Category: Salt Hydrolysis****COMMON EXAMINER TRAP / MISTAKE:**

Assuming sodium ethanoate (CH_3COONa) is neutral ($\text{pH} = 7$).

MODEL ANSWER / EXAMINER EXPECTATION:

Ethanoate ion hydrolyses in water: $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$, producing excess OH^- ions $\Rightarrow$ weakly alkaline solution ($\text{pH} \sim 8.9$).

FAQ #10: Mixing Acid and Base Neutralisation Volumes**Category: Total Volume Trap****COMMON EXAMINER TRAP / MISTAKE:**

Calculating final concentration using original volume instead of combined volume.

MODEL ANSWER / EXAMINER EXPECTATION:

Always add the two volumes together ($V_{\text{total}} = V_{\text{acid}} + V_{\text{base}}$) when calculating final concentrations of excess H^+ or OH^- .